卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0110 30-B003701S-Z0110

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

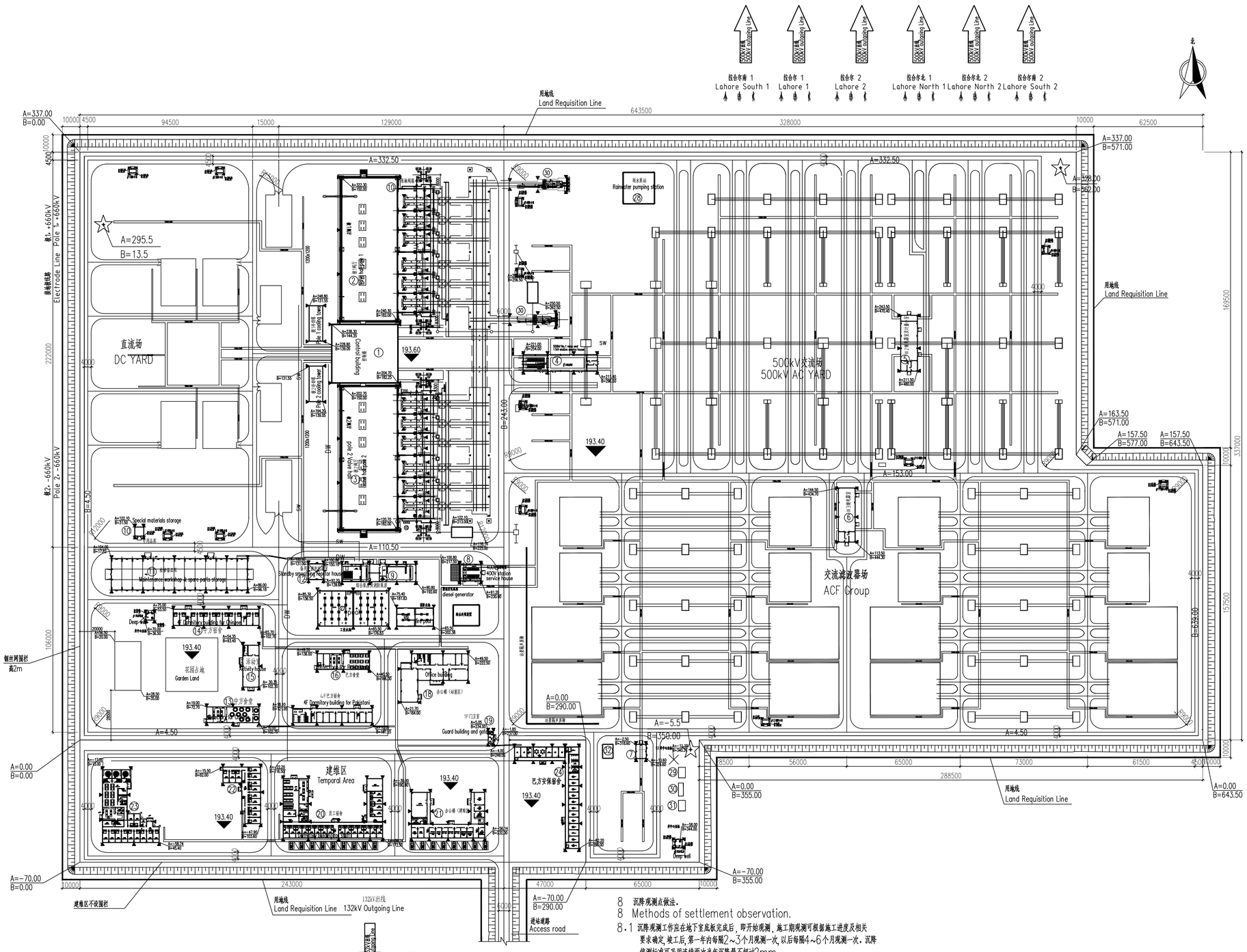
结构 专业 第 01 卷 第 10 册
Structure SPECIALITY VOLUME BOOK

卷册名称 全站沉降观测点布置图
Volume name Layout of observation point of total station settlement

批　准 APPROVED BY	
审　核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	
图纸 1 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

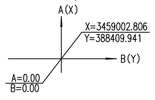
序号 No.	图　号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
2	30-B003701S-Z0110-001	全站沉降观测点布置图 Layout of observation point of total station settlement	A	1		

注:



沉降观测点布置图

说明:
Notes:
1.本图采用巴基斯坦UTM-43N平面坐标系,EGM2008高程系统。
1.Coordinate system: Pakistan UTM-43N. Elevation system: EGM2008
2.相对平面坐标以米(m)为单位,尺寸以毫米(mm)为单位。
2.Relative coordinates system is measured in meter. Dimensions are measured in millimeters.
3.该站平均设计标高为193.40m。
3.Site level is equal to 193.40m.
4.站内道路为环区型道路,道路中心标高193.60m。
4.Interior Road is suburban type, the center level of interior road is 193.60m
5.未注明的道路宽度均为3.0米,道路的转弯半径一般均为7.0米。
5.The width of interior roads which are not indicated is 3.0m. The average turning radius of interior roads is 7.0m
6.本图建筑(A,B)坐标与UTM-43N平面坐标系关系如下所示:
6.The relationship between relative coordinate system and UTM-43N system is as below:

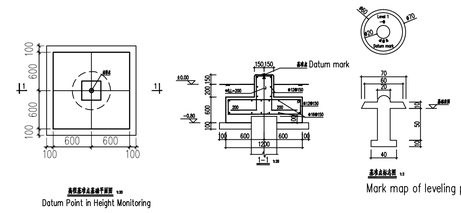


8 沉降观测点做法。
8 Methods of settlement observation.
8.1 沉降观测工作应在地下室底板完成后,即开始观测,施工期观测可根据施工进度及相关要求确定,竣工后,第一年内每隔2~3个月观测一次,以后每隔4~6个月观测一次。沉降停测标准可采用连续两次半年沉降量不超过2mm。
8.1 Settlement observation shall be started after completion of the basement floor, the observation can be determined according to the construction schedule and related requirements. After the final completion, the observation shall be made every 2~3 months for the first year , and shall be observed every 4~6 months afterwards. The settlement test standard can be that the settlement within six months shall not exceed 2mm for two consecutive times.
8.2 鉴于沉降观测资料连贯性的要求,严禁任意改用水准点和更改其标高。
8.2 In view of the consistency of settlement observation data, it is strictly forbidden to change the bench mark and the elevation.
8.3 ▲表示沉降观测点的平面位置,均设在柱或剪力墙上。
8.3 ▲represents the plane position of the settlement observation points, both on the column or shear wall.
9 本工程初设设计尚未最终审定,最终方案有存在调整可能,是否按图施工由总包CET决定,由此造成的变更设计不予负责。
8.3 As the preliminary design documents have not yet been approved finally, the project solution has probability to be modified. The design contractor has no responsibility for the project solution changed, if the construction contractor does not get authorized by CET before construction.

图例:
Legend:
建筑物 Building
道路 Road
围墙 Enclosing wall
地下建筑 Building underground
架构 Substaion frame
电缆沟 Cable trench
▲ 侧面式沉降观测点
■ 顶面式沉降观测点
☆ 高程基准点(永久沉降观测点)

建(构)筑物一览表
Buildings and structures list

编号 No.	名称 Name of buildings	沉降观测点个数 Settlement observation point	测量类型 Type of measurement	变形测量等级 Measurement grade	测量高度 height
1	控制楼 Control building	4	沉降观测 settlement observation	二级 Level 2	193.90
2	阀1厅户外跨度变基础 Valve hall pole 1	6+24	沉降观测 settlement observation	一级 Level 1	193.90
3	阀2厅户外跨度变基础 Valve hall pole 2	6+24	沉降观测 settlement observation	一级 Level 1	193.90
4	1厅继电器室及110/6kV站用变 500kV No.1 relay and 110kV station service power house	8	沉降观测 settlement observation	二级 Level 2	193.90
5	2厅继电器室及计量小室 500kV No.2 relay and metering house	6	沉降观测 settlement observation	二级 Level 2	193.90
6	3厅继电器室 500kV No.3 relay-house	6	沉降观测 settlement observation	二级 Level 2	193.90
7	132kV继电器室 132kV relay-house	4	沉降观测 settlement observation	二级 Level 2	193.90
8	400V站用电室 400V station service power house	4	沉降观测 settlement observation	二级 Level 2	193.90
9	综合泵房及消防泵房 Pump house	8	沉降观测 settlement observation	二级 Level 2	193.90
10	专用备库 Special materials storage	4	沉降观测 settlement observation	三级 Level 3	193.90
11	检修备品库 Maintenance workshop & spare parts storage	12	沉降观测 settlement observation	三级 Level 3	193.90
12	备用平波电抗器室 Standby smoothing reactor house	4	沉降观测 settlement observation	二级 Level 2	193.90
13	中方餐食 Refectory for Chinese	6	沉降观测 settlement observation	三级 Level 3	193.90
14	中方宿舍 Dormitory building for Chinese	8	沉降观测 settlement observation	三级 Level 3	193.90
15	活动室 Activity house	6	沉降观测 settlement observation	三级 Level 3	193.90
16	巴方食堂 Refectory for Pakistani	6	沉降观测 settlement observation	三级 Level 3	193.90
17	巴方宿舍 Dormitory building for Pakistani	8	沉降观测 settlement observation	三级 Level 3	193.90
18	办公楼(运维区) Office building (Operation and Maintenance Area)	8	沉降观测 settlement observation	三级 Level 3	193.90
19	门卫室 Guard building and gate	4	沉降观测 settlement observation	三级 Level 3	193.90
20	员工宿舍 Dormitory building for staff	18	沉降观测 settlement observation	三级 Level 3	193.90
21	办公楼(建维区) Office building (Temporal Area)	18	沉降观测 settlement observation	三级 Level 3	193.90
22	管理人员宿舍全部 Dormitory building for managers	9	沉降观测 settlement observation	三级 Level 3	193.90
23	管理人员宿舍半部 Dormitory building for managers	10	沉降观测 settlement observation	三级 Level 3	193.90
24	巴方安保宿舍 Dormitory building for security personnel	10	沉降观测 settlement observation	三级 Level 3	193.90
25	深井机房 Deep well room	2X4	沉降观测 settlement observation	二级 Level 2	193.90
26	消防小室 Fire equipment storage room	4x12	沉降观测 settlement observation	二级 Level 2	193.90
27	工业水池 pool	6	沉降观测 settlement observation	二级 Level 2	193.55
28	消防水池 fire pool	4	沉降观测 settlement observation	二级 Level 2	193.55
29	冷却塔 Pole 1/2 coding tower	2X4	沉降观测 settlement observation	二级 Level 2	193.55
30	备用变 Spare Converter Transformer	2X6	沉降观测 settlement observation	二级 Level 2	193.55
31	高程基准点 Datum Point in Height Monitoring	3	沉降观测 settlement observation	一级 Level 1	193.55
合计 Total		320			



Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND ENGINEERING CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - E N G - C E T - D D - G L - 0 1 0 1 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	闫	2018.11.20
校核 Checked	刘	2018.11.20
审核 Reviewed	马	2018.11.20
批准 Approved	杨	2018.11.20

全站沉降观测点布置图
Layout of observation point of total station settlement

比例 Scale: 1:1200 | 图号 DWG NO.: B003701S-Z0110-01 | 版本 REV.: A

A1(841x594)